

Section A - Physics

1. What is the SI unit of electric current?

- (A) Ampere
- (B) Volt
- (C) Ohm
- (D) Watt

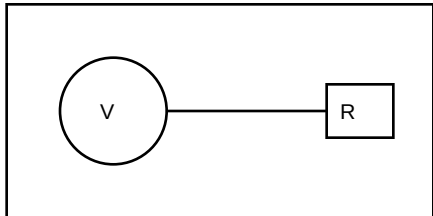
2. A body of mass 2 kg moves with acceleration 3 m/s^2 . Find the net force.

- (A) 3 N
- (B) 5 N
- (C) 6 N
- (D) 1.5 N

3. Light year is a unit of which physical quantity?

- (A) Time
- (B) Distance
- (C) Speed
- (D) Intensity

4. Refer to the circuit figure below. Identify the component labeled R.



- (A) Inductor
- (B) Capacitor
- (C) Resistor
- (D) Diode

5. Sound waves cannot travel through which of the following media?

- (A) Water
- (B) Steel
- (C) Air
- (D) Vacuum

Section B - Chemistry

Q.6 What is the chemical formula of water?

- (A) H₂O
- (B) CO₂
- (C) NaCl
- (D) CH₄

Q.7 Which gas is released during photosynthesis?

- (A) Nitrogen
- (B) Oxygen
- (C) Carbon dioxide
- (D) Hydrogen

Q.8 What is the pH value of pure water at 25 degrees Celsius?

- (A) 0
- (B) 14
- (C) 7
- (D) 1

Q.9 Refer to the table below comparing parameters across conditions.

Parameter	Value A	Value B
Voltage	220 V	110 V
Current	5 A	10 A

- (A) Condition A
- (B) Condition B
- (C) Both equal
- (D) None

Q.10 Which element is known as the King of Chemicals?

- (A) Sulphuric Acid
- (B) Hydrochloric Acid
- (C) Nitric Acid
- (D) Acetic Acid

Section C - Mathematics

11) Find the value of x if $2x + 6 = 14$.

- (A) 2
- (B) 3
- (C) 4
- (D) 5

12) What is the derivative of $\sin(x)$ with respect to x ?

- (A) $\cos(x)$
- (B) $-\cos(x)$
- (C) $\tan(x)$
- (D) $-\sin(x)$

13) What is the area of a circle with radius 7 cm? (Use $\pi = 22/7$)

- (A) 154 cm^2
- (B) 44 cm^2
- (C) 77 cm^2
- (D) 308 cm^2

14) If matrix A is 2×3 and matrix B is 3×2 , what is the order of matrix AB ?

- (A) 3×3
- (B) 2×2
- (C) 2×3
- (D) 3×2

15) What is the probability of getting a prime number when throwing a fair 6-sided die?

- (A) $1/6$
- (B) $1/3$
- (C) $1/2$
- (D) $2/3$

Section C - Mathematics (Contd.)

(16) Find the roots of the quadratic equation $x^2 - 5x + 6 = 0$.

- (A) 2 and 3
- (B) 1 and 6
- (C) -2 and -3
- (D) -1 and -6

(17) What is the sum of interior angles of a pentagon?

- (A) 180 deg
- (B) 360 deg
- (C) 540 deg
- (D) 720 deg

(18) Evaluate the limit of $\sin(x)/x$ as x approaches 0.

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

(19) What is the distance between points (0,0) and (3,4)?

- (A) 3
- (B) 4
- (C) 5
- (D) 7

(20) If $\log_{10}(x) = 2$, what is the value of x ?

- (A) 20
- (B) 100
- (C) 200
- (D) 1000